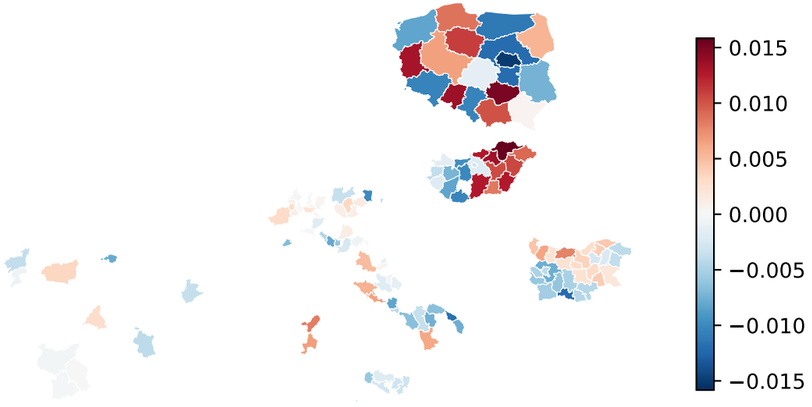
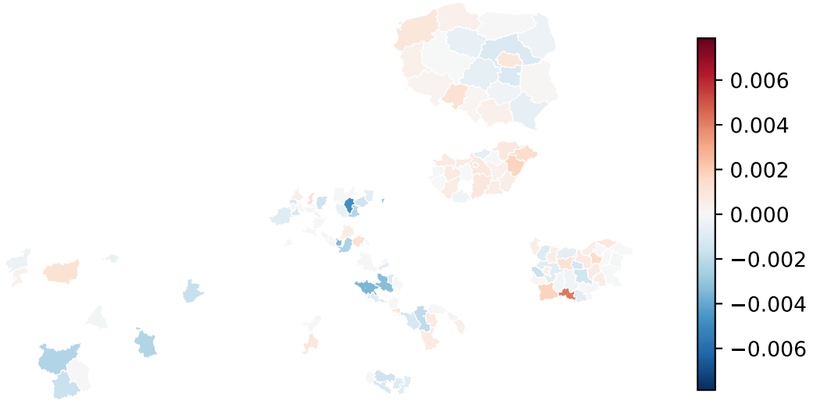
Energy-burden line: how each selected variable pushes households towards (red) or away from (blue) Double Risk, regional mean SHAP (CatBoost, pooled five-country model, stage B2)

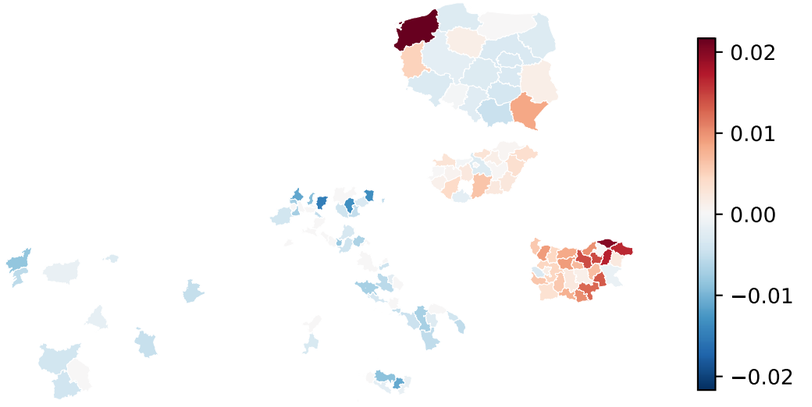
region (NUTS2)



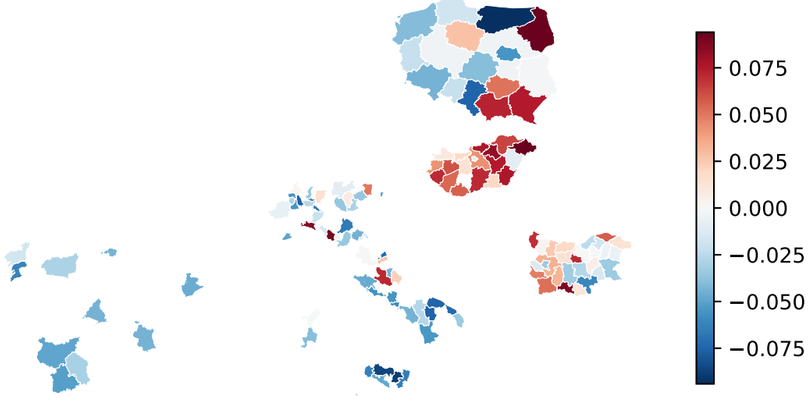
wealth index (house-value stand-in)



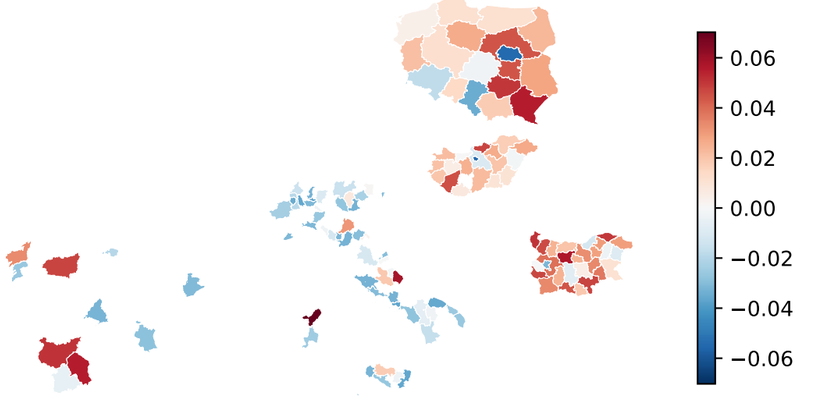
income per adult equivalent (EUR)



heating need (kWh)



density x urban area



solar radiation (winter)

